33kV OVERHEAD LINE FROM PARKSTON
POWER STATION POLE FIM FA35 RECLOSER
2000-F00-SLD-EL-10001

INTERTRIPS TO FIM FA35 RECLOSER

F12-TR-012
33 /11 kV
5000 kVA
Dyn11
z=7%
KNAN

F12-NR-012
PRIMARY CRUSHING TRANSFORMER
F12-TR-012 NEUTRAL EARTHING CUBICLE
254Ω +/-5%
30/0.05A
25A FOR 10s
SENSING
ER-15kV

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

F12-SR-001
11kV PRIMARY CRUSHING SWITCHROOM

F12-SW-012
PRIMARY CRUSHING 11kV SWITCHBOARD

NOTE 4

TEST

300/1A
2.5VA
Cl 0.2

PQM
SEL735

PR
SE-330AU

EARTH FAULT INDICATION

TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)

UPSTREAM TRIPS

300/1A
2.5VA
5P20

CLOSE
TRIP
630A

NOTE 2

50 50G 51 51G AFD 50N 51N
PR
SEL 751

49 63 71

F12SR001-LCS
E-SHUT
NOTE 5

A01
2000-F12-SCM-EL-10101

NOTE 2

NOTE 3

A03

11kV 1250A 3-E 25kA FOR 3s

A04
2000-F12-SCM-EL-10107

A05
2000-F12-SCM-EL-10109

A02
2000-F12-SCM-EL-10103

2A
630A

NOTE 2

VT
11kV/√3 / 0.11kV/√3 / 0.11kV/√3
15VA CLASS 0.2
20VA CLASS 3P

160E

100/1A
2.5VA
5P20

50/1A
1.5VA
10P10

TEST

VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT

200/1A
2.5VA
5P20

50/1A
1.5VA
10P10

50/1A
2.5VA
5P20

50/1A
1.5VA
10P10

ACTIVATION OF SWRM E-SHUT
DETECTION OF EARTH FAULT
DETECTION OF ARC FAULT

F12CG401-VS
PRIMARY CRUSHER 2
11/3.3kV 1000kW VSD

M
3~
600 kW
RTD

F12-CG-401
PRIMARY CRUSHER 2
2000-F12-SCM-EL-10105

F12-TR-001
PRIMARY CRUSHING
11/0.433kV TRANSFORMER

F12-SW-001
PRIMARY CRUSHING
415V MCC INCOMER

REFER DRG 2000-F12-SLD-EL-10201

11kV SPARE
FULLY EQUIPPED
-

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.
7. UPSTREAM 11kV FEEDER SHALL BE TRIPPED ON THE EARTH & ARC FAULT DETECTION OF THE DOWNSTREAM INCOMER ACB.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTE 2, 10
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
27	UNDER VOLTAGE
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
49	TRANSFORMER TEMPERATURE
63	TRANSFORMER PRESSURE / BUCCHOLZ
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 04/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

DRG No.	REFERENCES
2000-F12-DRG-EL-10001	F12-SR-001 11kV PRIMARY CRUSHING SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F12-SLD-EL-10201	F12-SW-001 PRIMARY CRUSHING 415V MCC SINGLE LINE DIAGRAM SHEET 1
2000-F12-SLD-EL-10051	F12-SW-012 PRIMARY CRUSHING 11kV SWITCHBOARD INTERLOCKING SINGLE LINE DIAGRAM
2000-F00-SLD-EL-10001	OVERALL SITEWIDE SINGLE LINE DIAGRAM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	07AUG25	FAT REDLINES INCORPORATED	[redacted]	[redacted]	[redacted]	MK	[redacted]
0	22OCT24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: PRIMARY CRUSHING
F12-SW-012 11kV SWITCHBOARD
SINGLE LINE DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F12-SLD-EL-10001	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELF12 - PRIMARY CRUSHING\2000-F12-SLD-EL-10001.DWG 13AUG25 11:50AM HOTSEAT04